

UrbanHeat AI

Al Khor area study, Qatar

Satellite indicators for urban-planning site visits

Team Al Zubarah

UAE Space Academy hackathon, 5 October 2026

Planning use case

A planner needs a shortlist of places to inspect for shade or planting.

The map shows observed surface temperature, green signal and mapped buildings for each cell.

A site visit must establish pedestrian activity, existing shade, ownership and feasibility.

No user interview, measured time saving or cooling benefit is claimed.

Open satellite data and mapped context

Sentinel-2: green-pixel indicators from 5, 15 and 30 September 2026.

Landsat: surface temperature from 14 and 30 September. The 6 September scene fails coverage QA.

WorldCover 2021: historical land-cover context. OpenStreetMap: contributed building footprints downloaded 5 October.

Thermal native detail is about 100 m, delivered on a 30 m grid. We report 300 m cells.

Al Khor study window and shortlist

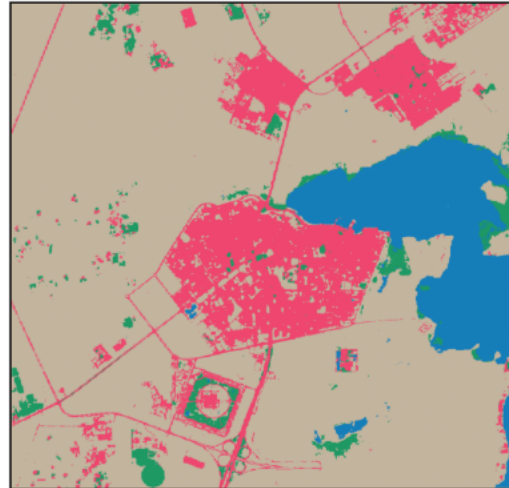
Al Khor · land-cover baseline and planning shortlist

Mapped building footprints · OSM



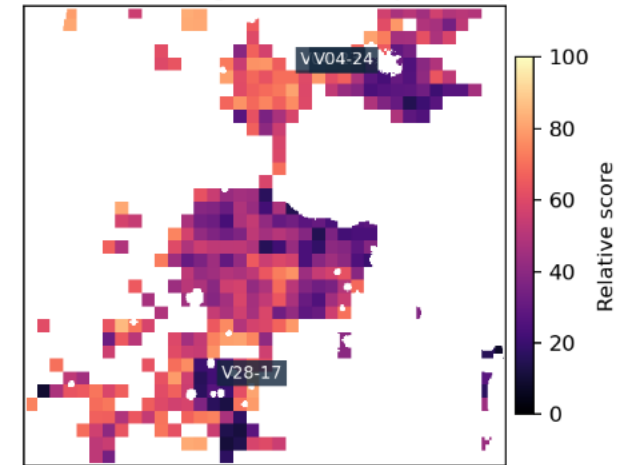
UTM east (m)

Historical land cover · WorldCover 2021



UTM east (m)

Relative investigation score · pilot only



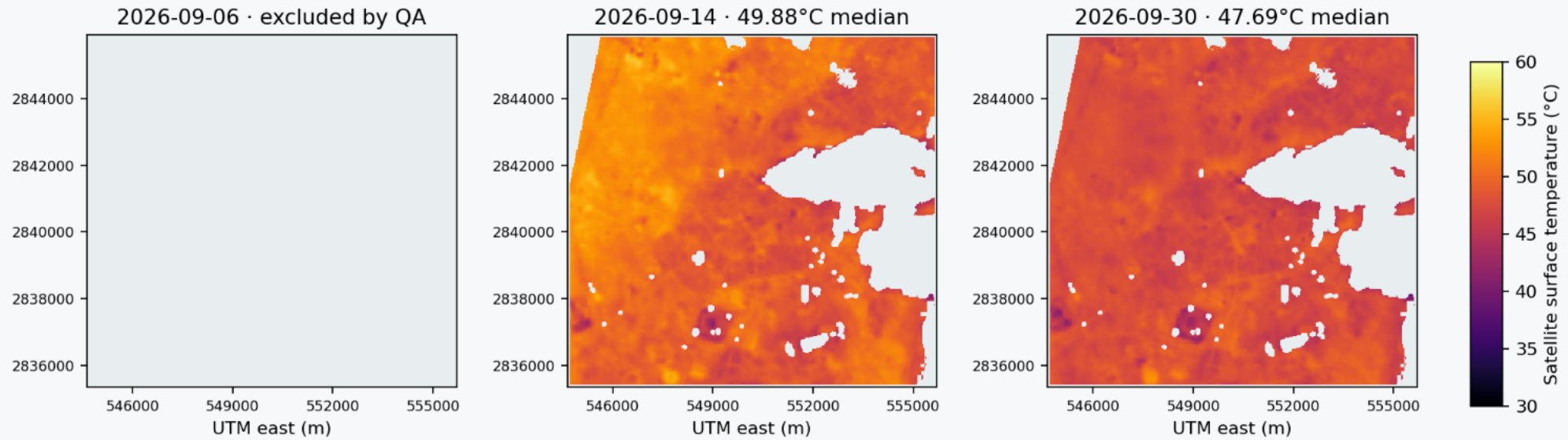
UTM east (m)

WorldCover: blue water, pink built-up, green vegetation, beige other · OSM footprints are contributed and may be incomplete
Shortlist requires site checks for pedestrian use, shade, ownership and planting feasibility. No health-risk or cooling benefit inferred.

1,111 observed cells and 373 screened urban candidates. Analyst-defined window, approximately 116 km².

Matched surface-temperature observations

Surface temperature · 3 dates checked · 2 pass the filters



Morning Landsat observations · 100 m thermal detail on a 30 m grid · Same colour scale
Water/coastal/uncertain pixels excluded; date-to-date differences are not a long-term warming trend or proof of vegetation cooling.

94.4379 km² of common inland sample footprints across the broad study window.

Leading candidates change with the settings

Candidate	Surface °C*	Green %*	Buildings %	Rank range
V04-23	50.4	0.0	2.3	1-14
V28-17	50.0	0.0	0.1	1-23
V04-24	50.2	0.1	2.9	2-34

Score: 50% temperature percentile, 30% inverse green percentile, 20% mapped-building percentile.

Thirteen sensitivity settings show no uniquely verified best intervention site.

*Means of date-level cell summaries. Rank ranges are scenario ranges, not confidence intervals.

Reproducibility and reference checks

The package includes 35 hashed input files, source metadata and pinned Python 3.12 dependencies.

Clean command-line regeneration passed. Normal Jupyter-kernel execution remains pending because this workspace blocks kernel sockets.

Grass reference agreement: 24/25. Building green-rule agreement: 0/30. Coastal points retained in thermal support: 0/30.

Contributed reference labels have conflicts. Human confirmation and independent thermal validation remain pending.

Prototype scope and next validation

The current prototype combines satellite indicators with contributed geometry and an upstream ML-derived land-cover product.

Our proposed VHR AI segmentation, paved-surface mapping and urban-expansion measurement remain future work.

Next: confirm the boundary and references, inspect candidate sites and assess suitable VHR data.

We have not established health risk, causal cooling, current shade coverage or long-term warming.

Team and review

Nasser Alhuri: registered team leader

Abdulrahman Almohannadi, Mohammed Almarri, Majed Alkuwari and Ali Alkubaisi

Actual contributions and other roles await team confirmation.

Before submission: complete registrations, confirm judge access and deadline, then approve the final package.

Code and writing were developed with AI assistance and require team review and ownership.

Data attribution and sources

Contains modified Copernicus Sentinel data (2026). Optical source: Element 84 Earth Search.

Landsat imagery courtesy of USGS. Collection 2 Level-2 surface temperature, doi:10.5066/P9OGBGM6.

ESA WorldCover 2021 v200, Zanaga et al. (2022). CC BY 4.0, doi:10.5281/zenodo.7254220.

© OpenStreetMap contributors. Building/reference geometry: ODbL 1.0. openstreetmap.org/copyright

Full scene IDs, exact provenance, processing rules and limitations accompany the package. Original code: MIT.